



IILM UNIVERSITY

GREATER NOIDA, UTTAR PRADESH

Thiranex- Skill Development & Future Tech

INTERNSHIP – C++ PROGRAMMING

C++ Programming

An Internship Experience in C++ Programming

INTERNSHIP DURATION

30 June 2026 — 29 July 2026

STUDENT CREDENTIALS

Khushi Jaiswal

Roll No.: 25SCS1003005486

Program: B.Tech
(CSE)

SUBMITTED TO

Ramvath Ganesh

Candidate & Internship Overview



Candidate Details

Student Name: Khushi Jaiswal
Roll No.: 25SCS1003005486
Degree: B.Tech (Computer Science & Engineering)
Batch: 2026–27
University: IILM University, Greater Noida, U.P.



Internship Profile

Organization: Thiranex – Skill Development & Future Tech
Domain: C++ Programming
Duration: 30 June 2026 – 29 July 2026
Mode: Project-based practical learning

Internship Focus

The internship focused on applying C++ concepts to functional console-based applications, with practical exposure to OOP, file handling, data management, loops, arrays, conditional statements and menu-driven programming.

Organization Profile: Thiranex

Thiranex – Skill Development & Future Tech

Thiranex provides practical, industry-oriented learning opportunities through project-based training and internship programs.

Internship Environment

- Project-based C++ programming tasks
- Functional console applications
- Coding, testing and debugging practice
- Bridge between academic concepts and practical implementation



Practical Learning

Hands-on assignments and real-world project exposure.



Skill Development

Strengthens technical knowledge, problem-solving and software development skills.



C++ Internship Focus

OOP, file handling, data management and console application development.

Problem Statement & Industry Context



The Challenge

Academic programming often emphasizes concepts and syntax, while students may have limited opportunities to apply them in practical software development. The internship addressed this gap through project-based C++ application development.



Solution Scope

Develop functional console-based applications by applying C++ fundamentals, OOP, file handling, data management, loops, arrays, conditions and menu-driven logic. Each project was designed, coded, tested and debugged.

Goal: bridge theoretical programming knowledge with practical application development.

Core Internship Objectives

1 Build Practical Confidence

Turn C++ theory into working programs through hands-on tasks.

2 Design Modular Programs

Break problems into functions, classes and manageable components.

3 Manage Real Records

Implement persistent student, customer and library records.

4 Develop User Interaction

Create clear menu-driven flows for common operations.

5 Strengthen Logic

Use conditions, loops and arrays to solve application-level problems.

6 Validate & Improve

Compile, test, debug and refine programs until functions work correctly.

Programming Stack & Implementation Areas

- **C++**

Primary programming language used throughout the internship.

- **Arrays & Data Handling**

Store and process structured information for application tasks.

- **File I/O**

Persistent storage and retrieval of records.

- **Testing & Debugging**

Compilation, execution, error tracing and functional checks.

- **Functions & Control Flow**

Functions, conditions, loops and menu logic.

- **Object-Oriented Programming**

Classes, objects, encapsulation, inheritance and polymorphism.

- **Console Applications**

Interactive, menu-driven interfaces for project operations.

- **Development Practice**

Requirement analysis → coding → testing → debugging → completion.

System Architecture

1 User Interface

Menu-driven console interface for user input and operation selection.

2 Input & Validation

Accepts input and performs basic validation.

3 Application Logic

Functions, loops, conditions, arrays and core program logic.

4 OOP Layer

Classes and objects organize data and functionality where required.

5 File Handling

Stores and retrieves records persistently.

6 Output Layer

Displays records, results, transactions and game boards.

Internship Timeline (Weeks 1 to 4)

1

Student Management System

Add, update, delete and display student records; file storage.

3

Library Management System

Book/member management, issue/return and search.

Week 2

2

Bank Management Application

Deposit, Withdrawal and balance checking using OOP and file handling.

Week 4

4

Mini Game Project

Tic Tac Toe / Snake using arrays, loops, conditions and game logic.

Key Projects & Demonstrated Skills



Student Management System

Record management • Add • Update • Delete • Display • File storage



Bank Management Application

Customer records • Deposit • Withdrawal • Balance inquiry • OOP



Library Management System

Books • Members • Issue/Return • Search by title or author



Mini Game Project

Game board • Player input • Loops • Arrays • Win/Loss detection • Replay

Collectively, these projects strengthened C++ programming, logical problem-solving, file handling and application development skills.

Project Implementation & Results

Development Flow



Implementation Result

All four assigned projects were successfully implemented as functional console applications, demonstrating practical use of C++ concepts and programming logic.



Skills Demonstrated

OOP • File handling • Data management • Logical problem-solving • Debugging • Menu-driven application development

Internship Offer & Engagement Details

1 Internship Role

Intern – C++ Programming

2 Intern ID

THX-JUN3026-435

3 Internship Dates

Commencement: 30 Jun 2026
Completion: 29 Jul 2026

4 Work Mode

Remote / Project-Based



Date: 30 June 2026

TO,
Khushi Jaiswal
Intern ID: THX-JUN3026-435

Subject: Offer Letter for Internship in C++ Programming

Dear **Khushi Jaiswal**,

Following our selection process, we are pleased to offer you an internship position at **Thiranex**. We believe your skills and enthusiasm will be a great addition to our team.

Internship Role	Intern - C++ Programming
Commencement Date	30 Jun 2026
Completion Date	29 Jul 2026
Work Mode	Remote / Project-Based

During this tenure, you will work on practical projects under industry mentorship. Your progress will be reviewed periodically, and your performance will determine the successful completion of the internship.

We look forward to a mutually beneficial learning journey. Please confirm your acceptance by starting your onboarding tasks.

This is an electronically generated document. No physical signature is required.
For verification, contact Thiranex Verification Cell.
www.thiranex.in



Offer Letter issued by Thiranex

Learning Outcomes

- Improved understanding of C++ fundamentals and practical implementation.
- Hands-on experience with OOP and file handling.
- Improved programming logic and problem-solving ability.
- Experience in developing functional console-based applications.
- Better understanding of testing, debugging and error handling.
- Confidence in applying academic programming knowledge to practical projects.

Academic concepts → Practical implementation → Stronger programming foundation

Conclusion

The C++ Programming Internship at Thiranex provided valuable practical exposure to programming and application development.

1

Practical Experience

Applied C++ concepts through four project-based applications.

2

Skill Development

Strengthened OOP, file handling, logic and debugging skills.

3

Future Foundation

Built a stronger foundation for further software development learning.

THANK YOU

Submitted to: Ramvath Ganesh